



Integrated Generalized Structured Component Analysis: On the Use of Model Fit Criteria in International Management Research

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Abstract

Structural equation modeling (SEM) has remained two mutually exclusive domains, factor-based vs. component-based, depending on whether a construct is modeled by either a factor or a component (i.e., weighted composite of indicators). Research in international management (IM) and international business (IB), however, needs to accommodate a more general model that considers a wide range of constructs from different disciplines at the same time, representing some constructs as factors (e.g., cultural distance and institutional distance) and others as components (e.g., international experience and export intensity). Integrated generalized structured component analysis (IGSCA) is a recently developed statistical method for estimating such models with both factors and components. IGSCA can provide overall fit indexes for model evaluation, including the goodness-of-fit index (GFI) and the standardized root mean squared residual (SRMR). However, the performance of these indexes in IGSCA is not yet investigated. Addressing this limitation, we (a) highlight the limitations of the dominantly used SEM approaches, (b) review the use of different SEM approaches in IM/IB research in the last decade, (c) conduct a simulation study, confirming that both GFI and SRMR distinguish well between correct and misspecified models with both factors and components, and (d) we illustrate the indexes' efficacy using a model concerning the role of personality traits and international experience in shaping cultural intelligence. Based on the review and the results of the simulation study and the illustrative example, we also propose rules-of-thumb cutoff criteria for each index in IGSCA.

Keywords Integrated generalized structured component analysis · Structural equation modeling · Model fit · GFI · SRMR · Cultural intelligence

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1 Introduction

International management (IM) and international business (IB) research is characterized by the complexity of the phenomena it investigates and their relationships (e.g., Eden & Nielsen, 2020). While some of the elements we study can be measured directly (i.e., observed variables), others can be measured only indirectly (i.e., constructs). Structural equation modeling (SEM) is a general multivariate method that can handle the inherent complexity of IM/IB phenomena by specifying and examining the intricate interrelationships between observed variables and constructs (e.g., Hult et al., 2006; Richter et al., 2016). With (common) factors and components, previous SEM research has used two statistical approaches to represent constructs derived from measurement theory. Depending on whether a construct is represented by a factor or a component, SEM has evolved into two domains – factor-based SEM and component-based SEM (e.g., Jöreskog & Wold, 1982; Rigdon, 2012; Rigdon and Sarstedt 2021; Tenenhaus, 2008). From a methodological perspective, covariance structure analysis (CSA; Jöreskog, 1970) is considered a standard statistical approach in factor-based SEM (Jöreskog, 1970). In contrast, generalized structured component analysis (GSCA; Hwang & Takane, 2004, 2014) and partial least squares path modeling (PLSPM; Lohmöller, 1989; Wold, 1982) are the most salient approaches used in component-based SEM (Hwang et al. 2020; Khan et al. 2019).

IM/IB researchers have increasingly realized that the two representations of constructs are theoretically incompatible (e.g., Beugelsdijk et al., 2018; Coltman et al., 2008; Richter et al., 2016) and that it is recommended using factor-based SEM approaches (e.g., CSA) to estimate models with factors while using component-based SEM approaches (e.g., GSCA) to estimate models with components. To ignore the match between construct representation and method would most likely produce biased estimates and model predictions, potentially leading to invalid inferences (Cho et al., 2021; Hair & Sarstedt, 2019; Hwang et al., 2021; Rigdon et al., 2017; Sarstedt et al., 2016). In order to test theoretical models that comprehensively explain IM/IB phenomena, researchers in this field often need to include both factors and components in the same model to analyze the pathways that lead to an outcome. For example, Obadia (2013) developed a component measure for foreignness-induced cognitive disorientation and assessed its nomological validity by estimating its impact on trust and export performance, both of which had been measured using factors. Similarly, Gilbert and Heinecke (2014) analyzed the impact of regional management autonomy and product/service adaptation on multinational corporations' regional success, drawing on a mix of components and factors. Further, IM/IB research is characterized by common control variables conceptualized by both components and factors (Cuervo-Cazurra et al., 2020; Nielsen & Raswant, 2018) and by using primary and secondary data combined, which often requires the joint use of components and factors (e.g., Cerar et al., 2021). IM/IB researchers, who frequently consider components and factors in their models, seem to be unaware of the methodological challenges brought on by accommodating both construct types in a single analysis. In fact, most SEM

approaches are ill-equipped for examining factors and components in the same model, and efforts to suitably extend them come with various challenges and limitations (e.g., Grace & Bollen, 2008; Sarstedt et al., 2016). Addressing this concern, Hwang et al. (2021) recently developed integrated generalized structured component analysis (IGSCA), a component-based approach to SEM that accommodates both construct types, while maintaining the flexibility of the original GSCA method. Due to being so recently introduced, IM/IB researchers have not integrated the IGSCA into their methodological toolbox. Nevertheless, it has obvious relevance for the field.

Thus, this paper's first objective is to enhance IM/IB researchers' awareness of how to correctly represent constructs and to introduce IGSCA as a novel analytical approach for handling such models. Importantly, IGSCA not only facilitates handling different kinds of constructs, but also comes with various overall fit measures that enable researchers to evaluate and compare different models. This sets the method apart from the more popular PLSPM method, which is restricted in terms of model fit assessment (Hair et al., 2019; Richter et al., 2016; Sarstedt et al., 2022). Research in the CSA domain has transferred well-known measures, such as the goodness-of-fit index (GFI; Jöreskog & Sörbom, 1986) and the standardized root mean square residual (SRMR), into a GSCA context. Cho et al. (2020) recently evaluated the efficacy of GFI and SRMR and derived cut-off values for their use in models with components only (i.e., GSCA). However, these findings and recommendations cannot readily be transferred to IGSCA, which considers a combination of components and factors. In light of the above, this study's second objective is to address this concern by conducting a simulation study in which we examine the measures' efficacy regarding differentiating between correct and incorrectly specified models that include both factors and components. Therefore, we advance understanding of IGSCA by evaluating the usefulness and applicability of two fit indices that are particularly useful in the assessment and comparative evaluation of models with different complexities. Such an assessment is of specific importance to IM/IB because this research area is characterized by the often-simultaneous use of factors and components (Richter et al., 2016), complex structural models (Eden & Nielsen., 2020), and conflicting theoretical frameworks and perspectives (Cuervo-Cazurra et al., 2020).

In pursuing these objectives, our study contributes to IM/IB research in two important ways. First, by calling attention to the correct representation of factors and components, we aim to increase researchers' awareness of the challenges of including components and factors in the same model without adequate analytical approaches. If IM/IB research does not recognize and respond to the often, complex mix of components and factors it could hinder the field's theoretical and practical advancement (Casson, 2018; Eden & Nielsen, 2020). IM/IB researchers who are not aware of the issues potentially arising in models that include both components and factors, might not analyze the data correctly, which could possibly result in biased findings and misleading theoretical and practical implications. In their recent review of methodological shortcomings in IM/IB research, Aguinis et al. (2020) identified deficient measures (i.e., the measure does not provide a complete understanding of the construct) and the multidimensionality of constructs in IM/IB research as

problematic, as the analysis can become a significant methodological challenge. As a solution to this specific problem, they have recommended using analytical techniques based on the conceptualization of a construct. In our article, we introduce IGSCA to the IM/IB field as an analytical approach that responds to such recent calls (e.g., Aguinis et al., 2020; Knight et al., 2021).

Our paper's second contribution, not limited to IM/IB research, is the assessment and evaluation of two fit indexes that can be used in IGSCA to indicate whether the proposed model reaches the minimum requirements based on the observed data. In doing so, we address recent calls to increase IM/IB research findings' reproducibility and trustworthiness (e.g., Aguinis et al., 2017; Cuervo-Cazurra et al., 2020; Delios, 2020; Norder et al., 2021). Specifically, the trustworthiness in empirical analysis can be improved by using methodological approaches that assure the IM/IB research findings are based on a rigorous, unbiased analysis (Cuervo-Cazurra et al., 2020) and by transparently reporting what potential limitations and biases the used methods have (Meyer et al., 2017). IGSCA's ability to use fit indexes enables researchers to evaluate theoretical model's possible consistency with the observed data and to compare alternative models, which is an important element of scientific research (Danks et al. 2020; Sharma et al. 2019, 2021). Our results suggest cut-off values for each of the indexes, which minimize Type I and Type II error rates under different conditions. We also illustrate the two indexes' application using a cultural intelligence model, which IM/IB researchers commonly use to assess a person's intercultural competence (e.g., Matsumoto & Hwang, 2013). Our empirical example not only illustrates the indexes' efficacy but also answers recent calls to improve our understanding of the role personality traits and international experience play in shaping cultural intelligence (Michailova & Ott, 2018; Richter et al., 2020).

The remainder of the paper is structured as follows: First, we provide a brief overview of factors and components, giving examples of their use in IM/IB research. We provide a brief review of different SEM approaches' capacity and shortcomings in handling factors and components in the same model and of these SEM approaches' use in IM/IB research in the last decade. Second, we report the results of a simulation study which investigated the use of cut-off values in assessing model fit in IGSCA. Third, we present the results of an empirical application of the fit indexes in an IM/IB context. Finally, we summarize the fit indexes' behaviors, suggest cut-off criteria, and discuss potential issues that researchers should consider when they utilize the indexes, with a particular emphasis on IM/IB studies.

2 On Using Models with Constructs in IM/IB Research

2.1 A Brief Overview and Examples of Factors and Components

SEM is a general multivariate method for specifying and estimating complex relationships between observed variables and constructs, represented by factors or components – as commonly hypothesized and empirically tested in IM/IB research (Eden & Nielsen, 2020; Hult et al., 2006; Richter et al., 2016). Figure 1 provides

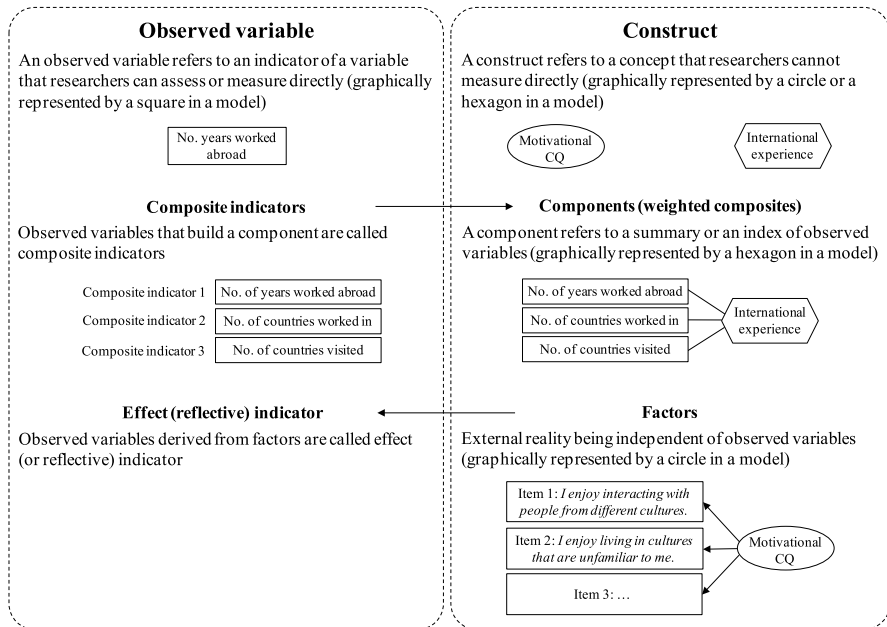


Fig. 1 An overview of observed variables, components, and factors

an overview of observed variables, constructs, and their relationships depending on whether we assume factors or components.

Observed variables refer to items and indicators of variables that are measured directly (e.g., Fornell & Larcker, 1981). In models they are graphically represented by a square. Constructs are mental abstractions that describe the ideas, objects, people, events, or phenomena researchers wish to measure or understand (Edwards & Bagozzi, 2000). With (common) factors and components, previous SEM research used two statistical approaches to represent constructs. On the one hand, if a researcher considers a construct to be an external reality independent of observed variables, which solely produces the correlation pattern of the observed variables, a factor can be used to represent this construct in a statistical model (Borsboom et al., 2004). Observed variables derived from factors are called effect (or reflective) indicators (Bollen, 1989; Edwards & Bagozzi, 2000; MacCallum & Browne, 1993). On the other hand, if a researcher theoretically defines a construct as a summary of observed variables or an index, so that the observed variables' correlational pattern forms the construct, a component or weighted composite of the observed variables can represent the construct in the model (e.g., Jarvis et al., 2003; Middendorp, 1991). Observed variables that build a component are commonly referred to as composite indicators (Bollen & Bauldry, 2011; Grace & Bollen, 2008).

Although IM/IB research has frequently used indexes (e.g., the cultural distance index and the institutional distance index), factors and components have rarely been discussed in previous research (Caprar et al., 2015; Cuypers et al., 2018; Kostova et al., 2020). Measurement theory guides which statistical representation is more appropriate

for a construct (Sarstedt et al., 2016). For example, Zeugner-Roth et al. (2008) conceptualize the construct country brand equity as a factor, which produces correlations among its measures, i.e., country brand loyalty, perceived country brand quality, and country brand awareness/associations. Similarly, constructs, such as international entrepreneurial orientation (Chen et al., 2020; Covin & Miller, 2014), international market orientation (Brännback et al., 2007; Knight & Kim, 2009), and institutional capabilities (Jean & Tan, 2019), are typically represented as factors. In contrast, constructs such as cross-cultural adjustment (Black & Stephens, 1989), global mindset (Gupta & Govindarajan, 2002), cultural intelligence (Ang et al., 2007), and foreignness-induced cognitive disorientation (Obadia, 2013) are often considered components, each measured by (a weighted) sum of observed variables that capture different dimensions or facets of the construct. Other constructs such as international performance (Hult et al., 2008; Richter et al., 2019) and export performance (Diamantopoulos, 1999; Sousa, 2004) are represented by either factors or components.

2.2 Different SEM Approaches and Their Ability to Handle Factors and Components

Depending on whether a construct is represented by a factor or a component, SEM has evolved into two domains – factor-based vs. component-based SEM (e.g., Jöreskog & Wold, 1982; Rigdon, 2012; Tenenhaus, 2008). From a methodological perspective, CSA is regarded as a standard statistical approach to factor-based SEM (Jöreskog, 1970), although other approaches are also available, which include the model-implied instrumental variable approach (MIIV; Bollen, 1996, 2019), consistent partial least squares (PLSc; Dijkstra, 2014; Dijkstra & Henseler, 2015), and generalized structured component analysis with measurement errors incorporated (GSCA_M; Hwang et al., 2017). Methods such as GSCA (Hwang & Takane, 2004, 2014) and PLSPM (Lohmöller, 1989; Wold, 1982) are the most flexible approaches to component-based SEM. These methods tend to produce similar estimates of parameters in unidimensional component models (Cho & Choi, 2020; Hair et al., 2017). Hwang et al. (2021) recently developed IGSCA, which incorporates GSCA and GSCA_M in a unified framework. This approach combines the methods for estimating components (GSCA) and factors (GSCA_M). Hwang et al. (2021) show that – different from other SEM methods – IGSCA produces unbiased parameter estimates in models that incorporate both factors and components. The method outperforms its component-based counterpart PLSc under most conditions of model specification, sample size, number of indicators per factor/component, and exogenous factor/component correlation. Adapted from Hwang et al. (2021), Table 1 summarizes the expected behaviors of different SEM approaches in estimating parameters of models with both factors and components.

2.3 A Brief Review of Different SEM Approaches Used in IM/IB from 2012 to 2021

Given that IM/IB phenomena often require researchers in this field to include factors and components in the same structural equation model and that only specific SEM approaches (i.e., IGSCA and PLSc) can handle both factors and components

Table 1 Summary of expected behaviors of SEM approaches in estimating parameters of models with both factors and components

SEM approach	Factors loading	Component loadings	Paths connecting factors only	Paths connecting components only	Paths connecting factors and components
CSA	0	–	0	+	+
PLSPM	+	0	–	0	–
GSCA _M	0	–	0	+	+
GSCA	+	0	–	0	–
PLSc	0	0	0	0	0
IGSCA	0	0	0	0	0

*SEM Structural Equation Modeling, *PLSPM* Partial Least Squares Path Modeling, *CSA* Covariance Structure Analysis, *PLSc* Consistent Partial Least Squares, *GSCA* Generalized Structured Component Analysis, *GSCA_M* Generalized Structured Component Analysis with Measurement Errors Incorporated, *IGSCA* Integrated Generalized Structured Component Analysis. 0 no bias, + = positive bias, and – = negative bias

*The table is adapted from Hwang et al., 2021

without bias, a question as to which SEM approaches IM/IB scholars have used, arises. Thus, in this section, we briefly review IM/IB studies that have used SEM approaches in the last decade (2012 until September 2021). We selected this time frame as both Richter et al. (2016) who reviewed the use of SEM in IM/IB research and Shackman (2013) who reviewed PLSPM and GSCA use in IM/IB research covered the time frame until 2013.

Following Tüselmann et al. (2016), we included the following six IM/IB journals in our literature review: *Management International Review*, *International Business Review*, *Journal of International Management*, *Journal of World Business*, *Journal of International Business Studies*, and *Global Strategy Journal*. We identified those articles that explicitly acknowledge having used a SEM approach for testing relationships between constructs. We excluded all studies that used SEM approaches solely for confirmatory factor analysis and measurement model assessment. We used different (combinations of) keywords to identify relevant studies in the *Web of Science*. This search process gave us a total of 203 articles, which were published in *Management International Review* (32), *International Business Review* (78), *Journal of International Management* (21), *Journal of World Business* (47), *Journal of International Business Studies* (20), and *Global Strategy Journal* (5). The number of articles using SEM varied from 31 in 2021 to 13 in 2015 and 2017. We found that a larger number of articles employ SEM in the decade our review covers than in previous decades (Richter et al., 2016). Thus, although the total number of articles published in IM/IB has grown across the last three decades, the relative use of SEM procedures remains relatively stable. We ascribe this finding to specific SEM software packages and general software packages' add-ons having become more accessible. Table 2 summarizes the articles, information on sample sizes, and the journals in which they were published over the past decade.

Table 2 SEM approaches and sample sizes in the major IM/IB outlets 2012 to 2021

SEM approach/sample size metrics	MIR	IBR	JIM	JWB	JIBS	GSJ	Total/Mean
PLSPM	13	39	14	15	4	1	86
CSA	19	37	7	32	16	4	115
PLSc	0	0	0	0	0	0	0
GSCA	0	2	0	0	0	0	2
GSCA _M	0	0	0	0	0	0	0
IGSCA	0	0	0	0	0	0	0
Total	32	78	21	47	20	5	203
Minimum <i>N</i>	42	65	68	63	85	87	68
Max <i>N</i>	29,987 (7,660)	1,600	2,175	1,668	5,919	228	3,208
Median <i>N</i>	206 (204)	170	208	210	248	186	204
Mean <i>N</i>	1,452 (638)	222	310	308	626	170	379
Standard deviation	5,131 (1,106)	207	430	350	1280	55	571

*SEM Structural Equation Modeling, *PLSPM* Partial Least Squares Path Modeling, *CSA* Covariance Structure Analysis, *PLSc* Consistent Partial Least Squares, *GSCA* Generalized Structured Component Analysis, *GSCA_M* Generalized Structured Component Analysis with Measurement Errors Incorporated, *IGSCA* Integrated Generalized Structured Component Analysis, *MIR* Management International Review, *IBR* International Business Review, *JIM* Journal of International Management, *JWB* Journal of World Business, *JIBS* Journal of International Business Studies, *GSJ* Global Strategy Journal. For *MIR* the sample sizes in brackets show the respective results excluding the Swoboda et al. (2017) study (*N* = 29,987)

Of the 203 papers, 115 used CSA (57%), 86 used PLSPM (42%), and two used GSCA (1%). None of the identified studies used PLSc, GSCA_M, or IGSCA, despite these methods' relevance for the research models being analyzed. For example, Lakshman et al. (2021) as well as Phookan and Sharma (2021) tested their structural model, including constructs that can be conceptualized as factors (e.g., cultural intelligence dimensions) and components (e.g., international experience and distance) based on regression analysis an CSA. Presbitero (2020) and Sharma (2019) used PLSPM to assess their respective structural model, including constructs that can be conceptualized as factors (e.g., cultural intelligence dimensions) and components (e.g., foreign language skill and international experience). While not all IM/IB studies include factors and components, factors and components are often used in the same structural model in IM/IB research. The observation that IM/IB researchers rely on CSA and PLSPM, is problematic as these approaches cannot include factors and components in the same model, which delivers potentially biased results and misleading implications. To the best of our knowledge, IM/IB researchers so far have used neither PLSc nor IGSCA, both of which are better able to model factors and components. This reflects a recent discussion in the field stating that IM/IB research often relies on standard analytical approaches and follows methodological dogmas instead of using methodological advancements and best practices (e.g., Aguinis et al., 2020; Casson, 2018; Nielsen et al., 2020; Poulis & Kastanakis, 2020; Yang et al., 2006). In their critical assessment of the current state of IB research, Poulis and Poulis (2018) noted that "...IB should be at the forefront of

deconstructing conceptualizations and resultant operationalizations that merely emulate unrealistic fit assumptions elsewhere. Instead, it unquestioningly embraces and reproduces them. The commonly found methodological choices in the field are in line with this isomorphic tendency. A covariance logic prevails and this accords with the rigid operationalization..." Given IGSCA's ability to model and estimate factors and components under different research conditions and its efficacy for model assessment, this approach is particularly suitable for extending IM/IB researchers' scope beyond such standard operationalizations.

3 Simulation Study

3.1 Design

We conduct a Monte-Carlo simulation study to assess whether the GFI and SRMR differentiate between the correct and misspecified models and to identify their cut-off values that minimize Types I and II error rates under different conditions. We explain how to compute the GFI and SRMR for the IGSCA model in the Appendix. In a Monte Carlo simulation study, researchers randomly generate a large number of samples under a pre-specified population model with given parameters and examine how well a statistical method can estimate the parameters or its goodness-of-fit indices perform over the samples (Paxton et al., 2001). As such Monte Carlo studies allow evaluating a method's performance from diverse perspectives under different scenarios and conditions (e.g., sample size, model specification, etc.), which is implausible to do with empirical data where parameters are typically unknown and many conditions are difficult to manipulate.

Cho et al. (2020) have conducted a simulation study to evaluate the efficacy of GFI and SRMR in GSCA that is developed for estimating models with components only. Conversely, our study considers more general models with both factors and components, which are frequently found in IM/IB studies as stated earlier. In addition, their experimental design was very concise, considering only one experimental factor (i.e., correlations between components), which may restrict the generalizability of their findings. To overcome the limitation, we consider a number of population models with both factors and components while manipulating four experimental factors: (1) the number of constructs, (2) the number of indicators per construct, (3) loading sizes, and (4) correlations between factors/components.

We consider three levels of the number of constructs ($N_{\text{construct}}=3, 5, \text{ and } 7$). We represent two constructs by factors and one construct by a component for $N_{\text{construct}}=3$, three constructs by factors and two constructs by components for $N_{\text{construct}}=5$, and four constructs by factors and three constructs by components for $N_{\text{construct}}=7$. Further, we consider three levels of the number of indicators per factor/component ($k=3, 5, \text{ and } 7$). In sum, we create 9 different model formulations (3 levels of the number of constructs $\times$ 3 levels of the numbers of indicators per construct), displayed in Fig. 2.

For each model formulation, we vary the values of loadings and correlations between constructs. We consider three loading sets in each level of the number of

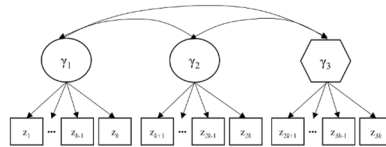
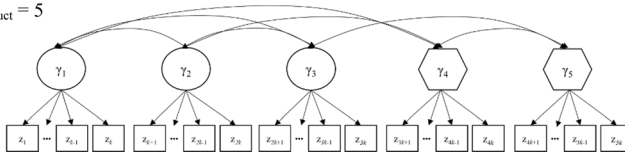
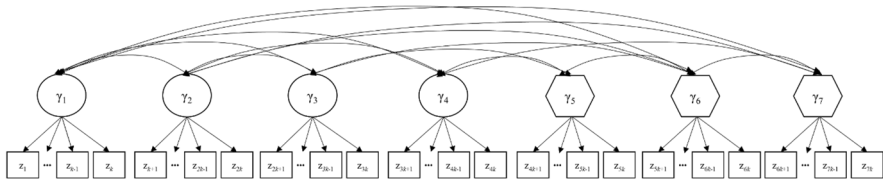
(a) For $N_{\text{construct}} = 3$ (b) For $N_{\text{construct}} = 5$ (c) For $N_{\text{construct}} = 7$ 

Fig. 2 Population models used in the simulation study. Circles represent factors, hexagons represent components, and squares represent indicators. Directional arrows denote loadings and bi-directional arrows represent correlations between factors/components. All the weights and residuals are omitted

indicators per construct. Specifically, we consider $\{0.5, 0.6, 0.7\}$, $\{0.6, 0.7, 0.8\}$, and $\{0.7, 0.8, 0.9\}$ for $k=3$, $\{0.5, 0.5, 0.6, 0.6, 0.7\}$, $\{0.6, 0.6, 0.7, 0.7, 0.8\}$, and $\{0.7, 0.7, 0.8, 0.8, 0.9\}$ for $k=5$, and $\{0.5, 0.5, 0.6, 0.6, 0.7, 0.7, 0.7\}$, $\{0.6, 0.6, 0.7, 0.7, 0.8, 0.8, 0.8\}$, and $\{0.7, 0.7, 0.8, 0.8, 0.9, 0.9, 0.9\}$ for $k=7$. We consider three levels of construct correlations ($r=0, 0.2$, and 0.4). In total, we contemplate 81 population models (9 model formulations $\times$ 3 loading sets $\times$ 3 construct correlations). The implied covariance matrix of each population model and its parameter values are given in Online Resources 1 and 2.

For each population model, we consider six different sample sizes ($N=50, 100, 200, 500, 1,000$, and $2,000$), for each of which we generate 1,000 random samples from a multivariate normal distribution with zero means and the implied covariance matrix derived is based on Hwang et al., (2021; see also Cho & Choi, 2020). We fit two models to each sample and compute their GFI and SRMR values. One model is the correct population model that generates the sample, as displayed in Fig. 2, whereas the other model is a misspecified model that incorrectly links a component or factor to indicators, as depicted in Fig. 3.

As in Hu and Bentler (1999), we assess a Type I error rate for a given cut-off value of the GFI by counting how frequently in the 1,000 samples per condition each fitted correct model's GFI value is *smaller* than the cut-off value, thus resulting in the correct model being rejected. Similarly, we compute a Type II error rate for the same cut-off value by counting how frequently each fitted misspecified model's GFI value is *greater* than the cut-off, in which case it fails to reject the misspecified model. Similarly, for the SRMR, we calculate the Type I and Type II error rates of a given cut-off

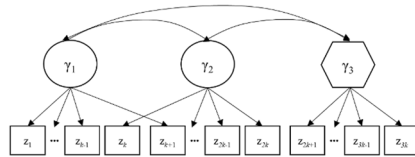
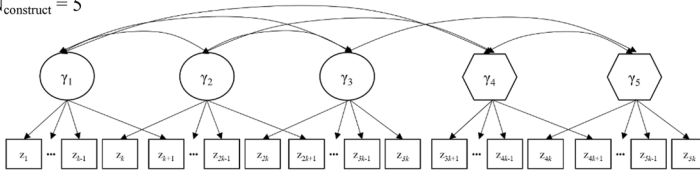
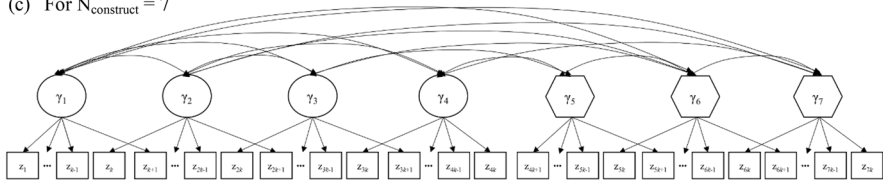
(a) For $N_{\text{construct}} = 3$ (b) For $N_{\text{construct}} = 5$ (c) For $N_{\text{construct}} = 7$ 

Fig. 3 Misspecified models used in the simulation study. Circles represent factors, hexagons represent components, and squares represent indicators. Directional arrows denote loadings and bi-directional arrows represent correlations between factors/components. All the weights and residuals are omitted

value by counting how frequently each fitted correct model's SRMR value is *greater* than the cut-off value and how frequently the fitted misspecified model's SRMR is *smaller* than the cut-off. Analogous to Cho et al. (2020), we compute the Type I and Type II error rates for the GFI and SRMR aggregated across the population models in each sample size. We give the Type I and Type II error rates for the GFI and SRMR for each population model per sample size in Online Resource 3.

3.2 Results

Table 3 presents the averages and standard deviations of the GFI and SRMR values for the correct and misspecified models in each sample size. As expected, the correct model's GFI value is large (e.g., > 0.93) except when the sample size is extremely small (i.e., $N=50$), approaching one when the sample size increases. At the same time, the misspecified model's GFI value, on average, is smaller than 0.83 in all sample sizes, although it tends to increase as the sample size increases. Further, except when $N=50$, the correct model's SRMR value is small overall (e.g., < 0.08) and approaches zero as the sample size increases. In general, the misspecified model's SRMR value is greater than 0.12 in all sample sizes, although it tends to be slightly smaller if the sample size has increased. In general, the clear differences in the average GFI and SRMR values between the correct and misspecified models suggest that these indices can readily be used for differentiating the two model specifications.

Table 3 Averages and standard deviations (SD) of the GFI and SRMR values of correct and incorrectly specified models aggregated over population models in each sample size

Sample size	Mean				SD			
	GFI		SRMR		GFI		SRMR	
	Correct	Misspecified	Correct	Misspecified	Correct	Misspecified	Correct	Misspecified
50	.886	.742	.018	.034	.102	.165	.018	.034
100	.937	.781	.010	.026	.071	.147	.010	.026
200	.966	.804	.005	.019	.051	.137	.005	.019
500	.986	.819	.002	.013	.032	.131	.002	.013
100	.993	.824	.001	.010	.023	.128	.001	.010
2,000	.996	.827	.001	.008	.016	.127	.001	.008

Table 4 shows the Types I and II error rates for different cut-off values of the GFI. As expected, a larger GFI cut-off value tends to produce a larger Type I error rate (i.e., more rejections of the correct model). At the same time, if the sample size increases, all the cut-off values lead to a Type I error rate close to or equal to zero. This is consistent with our previous finding that the correct model's average GFI values become large when the sample size increases (Table 3). Additionally, a larger GFI cut-off value tends to produce a smaller Type II error rate (i.e., more rejections of the misspecified model) across the sample sizes. Table 4 presents the average of Types I and II error rates for each GFI cut-off value, thereby showing which cut-off values result in the minimum average of both error rates. These results suggest that sample size should be considered when determining the cut-off values of the GFI.

Table 5 shows the Types I and II error rates for different SRMR cut-off values aggregated across the population models per sample size. As expected, a smaller SRMR cut-off value results in a larger Type I error rate in all sample sizes. When the sample size is very large ($N=2000$), all the cut-off values lead to zero Type I error rates. This is consistent with our previous finding that the correct model's average SRMR values are small in these sample sizes (Table 3). As expected, a smaller SRMR cut-off value results in a smaller Type II error rate. Table 5 then shows the average of Types I and II error rates for each SRMR cut-off value. These results again indicate that the provision of SRMR cut-off values is dependent on the sample size.

Figure 4 presents the averages of Type I and Type II error rates for different GFI and SRMR cut-off values across the sample sizes, which we calculated by averaging each cut-off value's mean error rates aggregated across the population models in each sample size. If $N=50$, a GFI cut-off value of 0.84 results in the minimum average of Types I and II error rates. As the sample size increases, larger GFI cut-off values lead to the minimum average Types I and II error rates, for instance, $GFI=0.87$ for $N=100$ and $GFI=0.91$ for $N=200$. If $N>200$, using a GFI cut-off value of 0.95 produces the minimum average of Type I and Type II error rates. Regarding the SRMR, using a cut-off value of 0.13 leads to the minimum average of Type I and Type II error rates for $N=50$. As the sample size becomes larger, smaller SRMR cut-off values are associated with the minimum average of both error rates, for instance, $SRMR=0.1$ for $N=100$ and $SRMR=0.08$ for $N=200$.

Table 4 Type I and Type II error rates and their averages for different cut-off values of GFI aggregated over population models in each sample size

Cut-off	N = 50			N = 100			N = 200			N = 500			N = 1,000			N = 2,000		
	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean
.80	.115	.287	.201	.000	.426	.213	.000	.531	.265	.000	.632	.316	.000	.660	.330	.000	.667	.334
.81	.143	.248	.196	.000	.387	.193	.000	.489	.245	.000	.577	.288	.000	.621	.311	.000	.631	.315
.82	.173	.210	.191	.002	.346	.174	.000	.448	.224	.000	.529	.265	.000	.573	.287	.000	.599	.300
.83	.203	.173	.188	.007	.301	.154	.000	.407	.204	.000	.485	.243	.000	.521	.261	.000	.551	.275
.84	.231	.141	.186	.018	.252	.135	.000	.358	.179	.000	.438	.219	.000	.469	.235	.000	.489	.244
.85	.262	.113	.187	.035	.203	.119	.000	.304	.152	.000	.383	.192	.000	.413	.207	.000	.425	.212
.86	.299	.088	.194	.051	.160	.106	.000	.249	.124	.000	.332	.166	.000	.365	.182	.000	.382	.191
.87	.343	.068	.206	.068	.124	.096	.000	.195	.097	.000	.276	.138	.000	.304	.152	.000	.319	.159
.88	.393	.052	.222	.095	.093	.094	.001	.148	.074	.000	.220	.110	.000	.253	.126	.000	.265	.132
.89	.447	.038	.243	.136	.068	.102	.001	.107	.054	.000	.162	.081	.000	.198	.099	.000	.216	.108
.90	.504	.026	.265	.187	.047	.117	.004	.075	.039	.000	.112	.056	.000	.139	.069	.000	.155	.078
.91	.566	.018	.292	.232	.029	.130	.018	.049	.033	.000	.074	.037	.000	.090	.045	.000	.100	.050
.92	.634	.011	.322	.280	.017	.148	.047	.027	.037	.000	.044	.022	.000	.054	.027	.000	.060	.030
.93	.707	.006	.356	.350	.008	.179	.076	.011	.044	.001	.021	.011	.000	.027	.013	.000	.030	.015
.94	.779	.003	.391	.440	.003	.222	.133	.003	.068	.002	.006	.004	.000	.009	.005	.000	.012	.006
.95	.846	.001	.423	.545	.001	.273	.218	.001	.109	.005	.001	.003	.000	.001	.000	.000	.000	.000

The bold cells indicate the minimum of the means of Type I and Type II error rates given the sample size

Table 5 Type I and Type II error rates and their averages for different cut-off values of SRMR aggregated over population models in each sample size

Cut-off	N = 50			N = 100			N = 200			N = 500			N = 1,000			N = 2,000		
	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean	Type I	Type II	Mean
.20	.000	.913	.457	.000	.962	.481	.000	.972	.486	.000	.975	.488	.000	.975	.488	.000	.975	.487
.19	.000	.839	.420	.000	.931	.465	.000	.955	.478	.000	.964	.482	.000	.966	.483	.000	.968	.484
.18	.000	.735	.367	.000	.870	.435	.000	.915	.458	.000	.935	.468	.000	.941	.471	.000	.941	.471
.17	.000	.616	.308	.000	.771	.386	.000	.846	.423	.000	.881	.441	.000	.889	.445	.000	.891	.446
.16	.000	.472	.236	.000	.676	.338	.000	.739	.370	.000	.772	.386	.000	.787	.394	.000	.796	.398
.15	.001	.296	.148	.000	.589	.294	.000	.654	.327	.000	.690	.345	.000	.706	.353	.000	.714	.357
.14	.004	.132	.068	.000	.459	.229	.000	.583	.292	.000	.624	.312	.000	.631	.316	.000	.634	.317
.13	.019	.050	.034	.000	.323	.162	.000	.465	.233	.000	.548	.274	.000	.567	.284	.000	.572	.286
.12	.117	.017	.067	.000	.185	.093	.000	.335	.168	.000	.423	.212	.000	.457	.229	.000	.475	.237
.11	.329	.005	.167	.003	.053	.028	.000	.243	.122	.000	.310	.155	.000	.331	.166	.000	.347	.173
.10	.556	.001	.278	.010	.012	.011	.001	.098	.049	.000	.235	.117	.000	.262	.131	.000	.268	.134
.09	.773	.000	.387	.034	.002	.018	.003	.016	.009	.000	.106	.053	.000	.162	.081	.000	.184	.092
.08	.912	.000	.456	.233	.000	.116	.009	.001	.005	.001	.011	.006	.000	.026	.013	.000	.044	.022
.07	.972	.000	.486	.556	.000	.278	.020	.000	.010	.002	.000	.001	.000	.001	.001	.000	.002	.001
.06	.993	.000	.496	.848	.000	.424	.099	.000	.050	.006	.000	.003	.001	.000	.000	.000	.000	.000
.05	.999	.000	.499	.967	.000	.483	.524	.000	.262	.013	.000	.007	.002	.000	.001	.000	.000	.000

The bold cells indicate the minimum of the average Type I and Type II error rates given the sample size

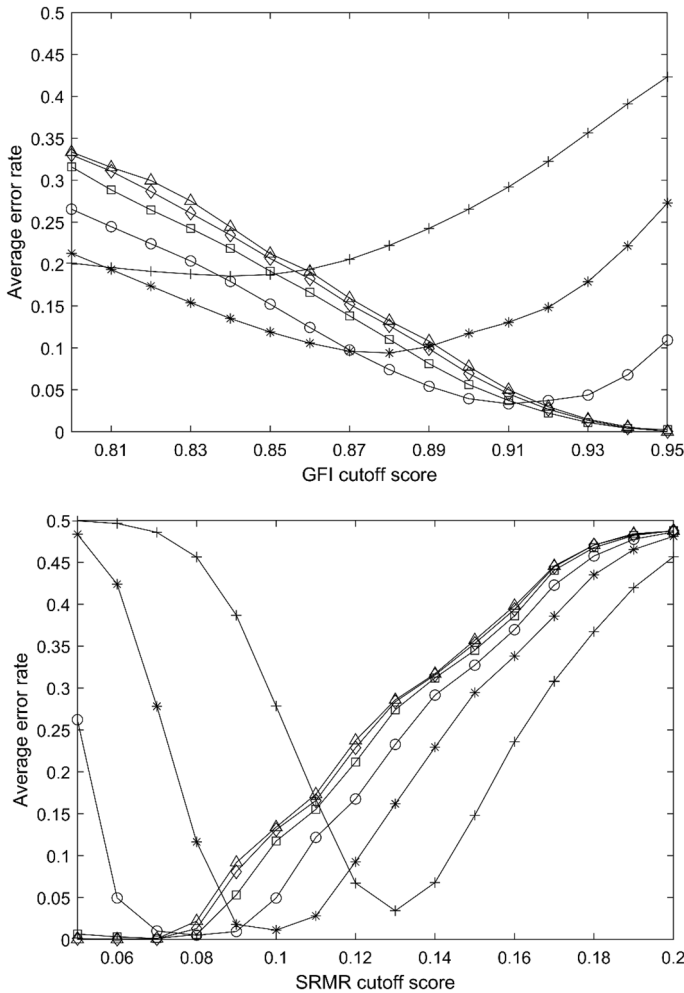


Fig. 4 Average error rates for different GFI and SRMR cut-off values aggregated over population models in each sample size. Symbols for representation of different sample sizes: + = 50, * = 100, ○ = 200, □ = 500, ◇ = 1,000, and Δ = 2,000

If $N > 200$, using an SRMR cut-off value of 0.06 produces the minimum average of error rates. Further, if $N \leq 200$, the minimum averages of Type I and II error rates led by SRMR cut-off values are always smaller than those led by GFI cut-off values. For example, if $N = 50$, the minimum average of the error rates associated with a GFI cut-off of 0.84 (i.e., 0.186) is much higher than that associated with an SRMR cut-off of 0.13 (i.e., 0.034). This indicates that using SRMR cut-off values should be preferred to using GFI cut-off values in detecting model misspecification in small samples. If $N > 200$, using GFI = 0.95 or SRMR = 0.06 provides the same minimum average (i.e., zero).

We also examine how the average Type I and Type II error rates in different combinations of GFI and SRMR cut-off values. As shown in Table 6, in respect

Table 6 Averages of Type I and Type II error rates across different combinations of cut-off values for GFI and SRMR aggregated over population models in each sample size

	<i>N</i>	Cut-off	SRMR																
			.20	.19	.18	.17	.16	.15	.14	.13	.12	.11	.10	.09	.08	.07	.06	.05	
GFI	50	.80	.199	.195	.186	.172	.154	.132	.108	.085	.087	.167	.278	.387	.456	.486	.496	.499	
		.81	.195	.192	.185	.174	.158	.139	.118	.098	.098	.168	.278	.387	.456	.486	.496	.499	
		.82	.191	.189	.184	.175	.163	.147	.129	.112	.110	.169	.278	.387	.456	.486	.496	.499	
		.83	.188	.187	.184	.177	.168	.155	.139	.125	.122	.171	.278	.387	.456	.486	.496	.499	
		.84	.186	.185	.183	.179	.172	.162	.149	.137	.134	.175	.278	.387	.456	.486	.496	.499	
	75	.85	.187	.187	.186	.183	.178	.170	.160	.150	.147	.180	.278	.387	.456	.486	.496	.499	
		.86	.194	.194	.193	.192	.188	.182	.175	.167	.163	.187	.279	.387	.456	.486	.496	.499	
		.87	.206	.206	.205	.204	.202	.198	.192	.186	.182	.198	.279	.387	.456	.486	.496	.499	
		.88	.222	.222	.222	.222	.220	.218	.213	.209	.206	.215	.280	.387	.456	.486	.496	.499	
		.89	.243	.243	.243	.242	.242	.240	.237	.234	.231	.237	.285	.387	.456	.486	.496	.499	
	100	.90	.265	.265	.265	.265	.265	.264	.262	.260	.258	.261	.294	.387	.456	.486	.496	.499	
		.91	.292	.292	.292	.292	.292	.292	.290	.289	.287	.288	.310	.387	.456	.486	.496	.499	
		.92	.322	.322	.322	.322	.322	.322	.322	.321	.320	.319	.331	.390	.456	.486	.496	.499	
		.93	.356	.356	.356	.356	.356	.356	.356	.356	.355	.355	.359	.396	.456	.486	.496	.499	
		.94	.391	.391	.391	.391	.391	.391	.391	.391	.391	.390	.392	.409	.457	.486	.496	.499	
	.95	.423	.423	.423	.423	.423	.423	.423	.423	.423	.423	.423	.423	.429	.458	.486	.496	.499	

Table 6 (continued)

<i>N</i>	Cut-off	SRMR																	
		.20	.19	.18	.17	.16	.15	.14	.13	.12	.11	.10	.09	.08	.07	.06	.05		
100	.80	.212	.210	.204	.191	.173	.151	.126	.094	.060	.027	.011	.018	.116	.278	.424	.483		
	.81	.193	.191	.187	.176	.161	.141	.117	.088	.056	.027	.011	.018	.116	.278	.424	.483		
	.82	.173	.172	.170	.161	.147	.130	.108	.083	.053	.026	.012	.018	.116	.278	.424	.483		
	.83	.154	.153	.152	.145	.134	.120	.101	.079	.052	.027	.014	.020	.116	.278	.424	.483		
	.84	.135	.135	.134	.130	.122	.111	.095	.076	.054	.031	.019	.025	.116	.278	.424	.483		
	.85	.119	.119	.118	.116	.111	.102	.090	.075	.058	.038	.027	.033	.116	.278	.424	.483		
	.86	.106	.106	.105	.104	.101	.094	.086	.073	.060	.045	.034	.039	.117	.278	.424	.483		
	.87	.096	.096	.096	.095	.093	.089	.082	.074	.062	.051	.042	.046	.117	.278	.424	.483		
	.88	.094	.094	.094	.094	.093	.090	.085	.079	.070	.062	.055	.057	.117	.278	.424	.483		
	.89	.102	.102	.102	.101	.101	.099	.096	.092	.086	.080	.074	.075	.120	.278	.424	.483		
	.90	.117	.117	.117	.117	.117	.116	.114	.111	.107	.103	.099	.099	.132	.278	.424	.483		
	.91	.130	.130	.130	.130	.130	.130	.129	.127	.125	.122	.120	.120	.145	.278	.424	.483		
	.92	.148	.148	.148	.148	.148	.148	.148	.147	.146	.144	.143	.143	.159	.278	.424	.483		
	.93	.179	.179	.179	.179	.179	.179	.179	.179	.178	.177	.176	.177	.186	.279	.424	.483		
	.94	.222	.222	.222	.222	.222	.222	.222	.222	.222	.221	.221	.221	.226	.285	.424	.483		
	.95	.273	.273	.273	.273	.273	.273	.273	.273	.273	.273	.273	.273	.274	.305	.424	.483		

Table 6 (continued)

<i>N</i>	Cut-off	SRMR																
		.20	.19	.18	.17	.16	.15	.14	.13	.12	.11	.10	.09	.08	.07	.06	.05	
200	.80	.265	.263	.258	.248	.231	.208	.186	.159	.125	.090	.040	.009	.005	.010	.050	.262	
	.81	.244	.243	.240	.232	.216	.195	.173	.148	.116	.084	.039	.009	.005	.010	.050	.262	
	.82	.224	.223	.221	.215	.201	.183	.162	.138	.108	.077	.038	.009	.005	.010	.050	.262	
	.83	.204	.203	.202	.197	.184	.169	.150	.128	.100	.071	.038	.009	.005	.010	.050	.262	
	.84	.179	.179	.178	.175	.166	.153	.137	.116	.093	.064	.036	.009	.005	.010	.050	.262	
	.85	.152	.152	.151	.150	.144	.135	.122	.104	.084	.059	.032	.009	.005	.010	.050	.262	
	.86	.124	.124	.124	.123	.120	.113	.104	.090	.074	.053	.029	.009	.005	.010	.050	.262	
	.87	.097	.097	.097	.097	.095	.090	.083	.074	.061	.046	.025	.009	.005	.010	.050	.262	
	.88	.074	.074	.074	.074	.073	.070	.064	.058	.048	.037	.023	.008	.005	.010	.050	.262	
	.89	.054	.054	.054	.054	.054	.052	.048	.043	.038	.028	.019	.008	.005	.010	.050	.262	
	.90	.039	.039	.039	.039	.039	.039	.036	.032	.029	.023	.017	.009	.006	.011	.050	.262	
	.91	.033	.033	.033	.033	.033	.033	.032	.030	.028	.024	.020	.015	.012	.017	.051	.262	
	.92	.037	.037	.037	.037	.037	.037	.037	.036	.035	.033	.031	.029	.026	.029	.058	.262	
	.93	.044	.044	.044	.044	.044	.044	.044	.044	.043	.042	.042	.041	.040	.042	.064	.262	
	.94	.068	.068	.068	.068	.068	.068	.068	.068	.068	.068	.068	.067	.067	.069	.082	.262	
	.95	.109	.109	.109	.109	.109	.109	.109	.109	.109	.109	.109	.109	.109	.110	.119	.262	

Table 6 (continued)

N	Cut-off	SRMR																	
		.20	.19	.18	.17	.16	.15	.14	.13	.12	.11	.10	.09	.08	.07	.06	.05		
500	.80	.315	.314	.309	.300	.284	.265	.242	.220	.187	.147	.112	.050	.006	.001	.003	.007		
	.81	.288	.287	.285	.278	.263	.245	.222	.200	.173	.136	.102	.046	.006	.001	.003	.007		
	.82	.265	.264	.263	.257	.243	.226	.204	.183	.157	.124	.091	.041	.006	.001	.003	.007		
	.83	.243	.242	.241	.238	.225	.211	.190	.169	.145	.114	.083	.038	.006	.001	.003	.007		
	.84	.219	.219	.218	.216	.205	.194	.177	.157	.135	.104	.076	.037	.006	.001	.003	.007		
	.85	.192	.192	.191	.189	.184	.175	.162	.144	.123	.095	.068	.037	.006	.001	.003	.007		
	.86	.166	.166	.166	.164	.161	.155	.145	.131	.112	.089	.062	.036	.006	.001	.003	.007		
	.87	.138	.138	.138	.137	.135	.131	.124	.114	.099	.081	.056	.034	.006	.001	.003	.007		
	.88	.110	.110	.110	.110	.109	.107	.101	.093	.083	.070	.050	.030	.006	.001	.003	.007		
	.89	.081	.081	.081	.081	.081	.080	.076	.069	.064	.053	.040	.024	.006	.001	.003	.007		
90	.90	.056	.056	.056	.056	.056	.056	.053	.048	.045	.037	.029	.019	.005	.001	.003	.007		
	.91	.037	.037	.037	.037	.037	.037	.037	.032	.030	.027	.019	.015	.004	.001	.003	.007		
	.92	.022	.022	.022	.022	.022	.022	.022	.021	.019	.019	.014	.011	.004	.001	.003	.007		
	.93	.011	.011	.011	.011	.011	.011	.011	.011	.010	.010	.009	.008	.004	.001	.003	.007		
	.94	.004	.004	.004	.004	.004	.004	.004	.004	.004	.004	.004	.004	.004	.002	.003	.007		
	.95	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.003	.007		

Table 6 (continued)

<i>N</i>	Cut-off	SRMR															
		.20	.19	.18	.17	.16	.15	.14	.13	.12	.11	.10	.09	.08	.07	.06	.05
1,000	.80	.329	.328	.324	.314	.300	.283	.258	.237	.206	.165	.131	.081	.013	.001	.000	.001
	.81	.311	.310	.309	.301	.286	.271	.246	.226	.200	.163	.129	.080	.013	.001	.000	.001
	.82	.287	.286	.286	.279	.266	.251	.227	.207	.184	.151	.118	.072	.013	.001	.000	.001
	.83	.261	.261	.260	.256	.243	.230	.207	.187	.165	.136	.104	.062	.013	.001	.000	.001
	.84	.235	.235	.234	.231	.221	.210	.191	.172	.151	.122	.092	.057	.013	.001	.000	.001
	.85	.207	.207	.206	.204	.200	.191	.177	.160	.139	.112	.083	.054	.013	.001	.000	.001
	.86	.182	.182	.182	.180	.177	.170	.161	.146	.127	.104	.075	.052	.013	.001	.000	.001
	.87	.152	.152	.152	.151	.149	.145	.138	.127	.112	.096	.069	.048	.013	.001	.000	.001
	.88	.126	.126	.126	.126	.125	.123	.117	.108	.099	.085	.063	.043	.013	.001	.000	.001
	.89	.099	.099	.099	.099	.099	.098	.094	.085	.082	.071	.055	.037	.013	.001	.000	.001
	.90	.069	.069	.069	.069	.069	.069	.067	.060	.058	.051	.040	.029	.011	.001	.000	.001
	.91	.045	.045	.045	.045	.045	.045	.045	.040	.038	.035	.025	.022	.009	.001	.000	.001
	.92	.027	.027	.027	.027	.027	.027	.027	.025	.024	.024	.017	.015	.007	.000	.000	.001
	.93	.013	.013	.013	.013	.013	.013	.013	.013	.013	.013	.012	.011	.006	.000	.000	.001
	.94	.005	.005	.005	.005	.005	.005	.005	.005	.005	.005	.005	.004	.004	.000	.000	.001
	.95	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.001

Table 6 (continued)

<i>N</i>	Cut-off	SRMR																		
		.20	.19	.18	.17	.16	.15	.14	.13	.12	.11	.10	.09	.08	.07	.06	.05			
2,000	.80	.333	.332	.328	.317	.304	.288	.263	.240	.212	.172	.134	.092	.022	.001	.000	.000			
	.81	.315	.315	.315	.306	.293	.279	.253	.231	.209	.172	.134	.092	.022	.001	.000	.000			
	.82	.300	.300	.299	.292	.279	.266	.240	.218	.198	.168	.130	.090	.022	.001	.000	.000			
	.83	.275	.275	.275	.270	.257	.246	.221	.199	.180	.153	.116	.080	.021	.001	.000	.000			
	.84	.244	.244	.244	.241	.231	.220	.201	.179	.161	.134	.100	.069	.021	.001	.000	.000			
	.85	.212	.212	.212	.209	.205	.198	.182	.164	.146	.121	.088	.062	.021	.001	.000	.000			
	.86	.191	.191	.191	.188	.185	.179	.170	.153	.136	.115	.081	.059	.021	.001	.000	.000			
	.87	.159	.159	.159	.158	.155	.152	.145	.132	.120	.105	.073	.053	.021	.001	.000	.000			
	.88	.132	.132	.132	.132	.130	.130	.124	.113	.105	.092	.068	.048	.021	.001	.000	.000			
	.89	.108	.108	.108	.108	.108	.107	.103	.093	.091	.080	.062	.043	.021	.001	.000	.000			
	.90	.078	.078	.078	.078	.078	.077	.075	.067	.066	.059	.047	.035	.018	.001	.000	.000			
	.91	.050	.050	.050	.050	.050	.050	.050	.044	.043	.040	.028	.026	.013	.001	.000	.000			
	.92	.030	.030	.030	.030	.030	.030	.030	.028	.027	.027	.018	.017	.011	.001	.000	.000			
	.93	.015	.015	.015	.015	.015	.015	.015	.015	.015	.015	.013	.012	.008	.000	.000	.000			
	.94	.006	.006	.006	.006	.006	.006	.006	.006	.006	.006	.006	.006	.005	.000	.000	.000			
	.95	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000			

The bold cells indicate the minimum of the average Type I and Type II error rates given the sample size

of $N \leq 500$, using an SRMR cut-off value in combination with a GFI cut-off value does not result in a smaller minimum average error rate than that obtained by using an SRMR cut-off value only. Conversely, when $N > 500$, using a combination of $\text{SRMR} = 0.06$ and $\text{GFI} = 0.95$ yields the same average error rate (i.e., zero) as using them separately. In this case, using an SRMR cut-off value of 0.06 seems to minimize the average value of Type I and Type II error rates, no matter which GFI cut-off value is chosen. Similarly, using a GFI cut-off value of 0.95 results in the minimum average of Types I and II error rates, regardless of the SRMR cut-off values.

4 Empirical Illustration

4.1 Model

To illustrate the efficacy of the model fit indexes in IGSCA, we draw on a model for cultural intelligence (CQ), which describes an individual's capability to adapt effectively to new cultural environments (Earley & Ang, 2003). CQ is commonly used in international management research (e.g., Yari et al., 2020) to assess a person's intercultural competence (e.g., Matsumoto & Hwang, 2013; Richter et al., 2020). Meta-analytic studies have shown that CQ is well-suited to predict work-related outcomes such as cross-cultural adjustment and job performance (Rockstuhl & Van Dyne, 2018; Schlaegel et al., 2017).

Research suggests that personality and biographical information, such as international experience, play an important role in developing intercultural competence

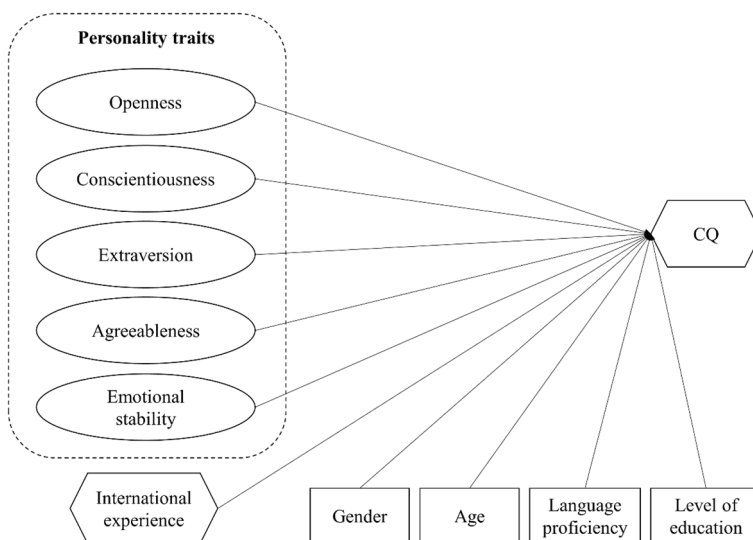


Fig. 5 Cultural intelligence (CQ) model. Circles represent factors, hexagons represent components, and squares represent control variables

generally (Leung et al., 2014) and of CQ in particular (Ang & van Dyne, 2008). Based on personality theory and experiential learning theory (Kolb, 1984), we therefore propose that the five-factor model's (FFM's) personality traits (i.e., openness, conscientiousness, extraversion, agreeableness, and emotional stability) and international experience are positively associated with CQ (Fig. 5). For example, personality theory suggests that individuals more open to experience are more curious and attentive to the environment surrounding them, are willing to try new things, and are more flexible and creative in their approach toward novel situations (e.g., Woo et al., 2014). Thus, open-minded individuals are more likely to develop the capacity to effectively function in a different cultural environment than are more close-minded individuals (e.g., Oolders et al., 2008). Relatedly, from an experiential learning theory perspective, learning occurs by reflection that transforms experience. When individuals reflect on the experience they gain in an international context, they can learn from that experience, thereby developing a higher CQ (e.g., Michailova & Ott, 2018).

4.2 Sample and Data Collection

We tested the model using data collected in 2019 from management students at Bachelor's and Master's level at a Dutch university. Specifically, we use two samples of international students from Belgium ($n=85$) and Germany ($n=195$). We used an online survey, to ensure consistency across samples with respect to the data collection procedure, as well as the survey format (Leung, 2008). Participants' mean age was 21.3 in the Belgian sample (44% were female and on average had completed a bachelor's degree) and 21.4 in the German sample (50% were female and on average were third year Bachelor students).

Although the management literature controversially discusses student samples as data (e.g., Peterson & Merunka, 2014), drawing on a sample of international students allowed us to examine CQ development during the time the respondents actually are gaining experience in a foreign environment (Bello et al., 2009). In addition, our main objective is to provide an empirical illustration of how model fit indices in IGSCA are used as example to guide IM/IB researchers. The sample is well suited to this purpose.

4.3 Measures

Cultural intelligence: The dependent variable, CQ, is measured with Van Dyne et al.'s (2012) extended CQ scale (E-CQS), which comprises four dimensions assumed to form a person's overall CQ (i.e., motivational CQ, cognitive CQ, behavioral CQ, and metacognitive CQ). Compared to the original 20-item CQ scale (Ang et al., 2007), this 37-item scale has shown evidence of higher

predictive power (Richter, Schlaegel, et al., 2020; Richter, van Bakel, et al., 2020). To the best of our knowledge, only a few studies have examined the CQ determinants measured with this extended scale. We conceptualize CQ as a higher-order component with the lower-order components motivational CQ, cognitive CQ, behavioral CQ, and metacognitive CQ jointly forming the higher-order component. We computed each construct by averaging the dimension-specific items as the efficacy of different item weighting methods for higher-order components has not been discussed in an IGSCA context. Specifically, while guidelines for the use of the (extended) repeated indicators and two-stage approaches in a PLSPM context exist, there is no corresponding research for IGSCA. In addition, using equal indicator weights is the approach commonly used in this stream of research (e.g., Schlaegel et al., 2021) and related research fields (Sarstedt et al., 2022). All items are measured on a seven-point Likert scale, ranging from 1 = strongly disagree to 7 = strongly agree.¹

We used the short form of the International Personality Item Pool (IPIP) to measure the FFM *personality traits*. The Mini IPIP (Donnellan et al., 2006) is a 20-item short version of the original 50-item measure. Each trait is measured with five items on a seven-point Likert scale, ranging from 1 = strongly disagree to 7 = strongly agree. For *international experience*, we use a composite measure of international experience drawing on six indicators that cover breadth, depth, and intensity of international experience (Maddux et al., 2020). To measure each aspect, the respondents first mentioned the number of foreign countries in which they had lived for more than three months. Second, we asked about the number of countries in which the respondent had received formal education. Third, we asked about the number of countries they have visited. Fourth, respondents estimated the number of months they had spent outside their home country. Finally, they indicated how much time they had spent in the foreign country in which they had stayed the longest uninterruptedly.

Drawing on prior research (Nielsen & Raswant, 2018), our analysis considers age, gender, language proficiency, and level of education as *control variables*. The gender question includes three options: female, male, and other. As no respondents selected 'other,' gender was measured with a dummy variable (female = 1, male = 0). We measured language proficiency based on the number of foreign languages the respondents reported themselves to know at least at a conversational level. Finally, we measured the level of education by a single question with nine options (1 = high school graduate or less, 2 = first year Bachelor, 3 = second year Bachelor, 4 = third year Bachelor, 5 = Bachelor degree or equivalent, 6 = first year Master, 7 = second year Master, 8 = Master degree of equivalent, 9 = PhD or equivalent). In addition to the control variables, we also asked about respondents' personal background information on nationality and current country of residence.

¹ The E-CQS scale is copyright-protected, but is available on request at cquery@culturalq.com.

4.4 Results

We applied IGSCA to estimate the model. Using 1,000 bootstrap samples we calculated the model estimates' 95% confidence intervals. Table 7 shows the measurement model's evaluation. For the German sample ($n=195$), we find that all factor measures exhibit satisfactory levels of internal consistency reliability in having Cronbach's alpha and composite reliability values above the recommended threshold

Table 7 Measurement model evaluation results

	Item	GERMANY (<i>N</i> =195)				BELGIUM (<i>N</i> =85)			
		Loading	AVE	Cronbach's alpha	CR	Loading	AVE	Cronbach's alpha	CR
<i>Factors</i>									
Openness	<i>open1</i>	.617	.399	.689	.722	.852	.435	.675	.725
	<i>open2</i>	.499				.365			
	<i>open3</i>	.619				.356			
	<i>open4</i>	.763				.868			
Conscientious- ness	<i>cons1</i>	.671	.500	.731	.750	.740	.400	.617	.663
	<i>cons2</i>	.677				.543			
	<i>cons3</i>	.769				.600			
Extraversion	<i>extra1</i>	.693	.503	.789	.802	.449	.427	.699	.740
	<i>extra2</i>	.698				.808			
	<i>extra3</i>	.704				.712			
	<i>extra4</i>	.741				.587			
Agreeableness	<i>agre1</i>	.781	.428	.727	.747	.694	.404	.682	.721
	<i>agre2</i>	.610				.588			
	<i>agre3</i>	.615				.779			
	<i>agre4</i>	.594				.425			
Emotional stability	<i>emot1</i>	.699	.499	.712	.744	.818	.450	.653	.704
	<i>emot2</i>	.556				.628			
	<i>emot3</i>	.837				.537			
Composites	Item	Loadings	Weight	VIF		Loadings	Weight	VIF	
International experience	<i>inte1</i>	.595	.278	1.246		.641	.274	1.236	
	<i>inte2</i>	.737	.352	1.354		.735	.312	1.398	
	<i>inte3</i>	.628	.306	1.231		.315	.132	1.048	
	<i>inte4</i>	.710	.343	1.452		.825	.345	1.799	
	<i>inte5</i>	.542	.258	1.314		.798	.339	1.704	
CQ	<i>cq1</i>	.799	.327	1.632		.825	.400	1.576	
	<i>cq2</i>	.812	.335	1.704		.788	.358	1.514	
	<i>cq3</i>	.742	.330	1.380		.685	.315	1.333	
	<i>cq4</i>	.770	.287	1.590		.659	.261	1.382	

Note: All loadings and weights are significant ($p < .05$), except *inte3* in the Belgium sample. AVE=average variance extracted; CR=composite reliability; VIF=variance inflation factor

Table 8 HTMT values

	Openness	Conscientiousness	Extraversion	Agreeableness	Emotional stability
Openness		.089	.146	.284	.246
Conscientiousness	.317		.064	.090	.163
Extraversion	.011	.255		.360	.082
Agreeableness	.146	.164	.624		.077
Emotional stability	.225	.215	.408	.121	

Upper diagonal = Germany sample; lower diagonal = Belgium sample

of 0.7 (Hair et al. 2022). The only exception is in the measurement of openness, which has a slightly lower Cronbach's alpha value (0.689). Loadings are generally high (>0.7) except for some indicators of openness and agreeableness, resulting in diminished levels of convergent validity below 0.5. However, taking the context of this study into account, these slight deviations are not considered problematic. As Table 8 shows, all measures exhibit discriminant validity, as heterotrait-monotrait (HTMT) values considerably lower than the (conservative) threshold of 0.85 confirm (Henseler et al., 2015). In analyzing the composite measures of international experience and CQ, we find that the indicators are not subject to collinearity issues, as indicated by VIF values below 3 (Table 7). The bootstrapping results show that all indicator weights are significant, which supports their empirical relevance (Hair et al. 2022).

Table 7 also shows the measurement model evaluation results for the Belgium sample ($n=85$). Compared to the German sample, all factor measures exhibit lower levels of internal consistency reliability and convergent validity. For example, the conscientiousness measure does not meet the common thresholds assumed for internal consistency reliability and convergent validity. In contrast, all HTMT values are low (Table 8), which supports the measures' discriminant validity. The composite measures generally exhibit satisfactory psychometric properties. All VIF values are low and most indicator weights are significant (Table 7). Only one indicator of international experience has a non-significant weight (i.e., the number of countries in which the respondent received formal education). As the same indicator's loading is also low (0.315) and non-significant, this indicator is neither relatively, nor absolutely relevant for measuring international experience.

The measurement model's quality is also mirrored in the overall model fit (Table 9). Specifically, the German sample produces a GFI value of 0.904, which is slightly lower than our recommended cut-off if the sample size is around 200 (0.91). At the same time, however, the SRMR value of 0.077 is clearly lower than our recommended cut-off (0.08), which indicates a good model fit. We therefore conclude that the model exhibits an acceptable fit for the German sample. Conversely, the results show that the same model clearly lacks fit when we use the Belgium sample for its estimation. The GFI value (0.832) is much lower than the cut-off assumed for a sample size of around 100 (0.88). Similarly, the analysis yields an SRMR value

Table 9 Structural model results

Path relationship	GERMANY (<i>N</i> =195)		BELGIUM (<i>N</i> =85)	
	Estimate	95% confidence interval	Estimate	95% confidence interval
Openness	.165	[−.042; .353]	.215	[−.066; .453]
Conscientiousness	.046	[−.109; .205]	.114	[−.254; .454]
Extraversion	.105	[−.070; .279]	−.059	[−.484; .420]
Agreeableness	.236	[.035; .443]	.226	[−.135; .593]
Emotional stability	−.091	[−.299; .111]	−.148	[−.446; .173]
International experience	.106	[−.027; .270]	.088	[−.177; .405]
Age	.097	[−.061; .230]	−.082	[−.454; .251]
Gender	−.063	[−.286; .116]	−.180	[−.441; .080]
Language proficiency	.144	[−.001; .279]	.248	[−.029; .493]
Education	.097	[−.087; .250]	.144	[−.133; .474]
GFI	.904		.832	
SRMR	.077		.104	

(0.104), which is slightly higher than the corresponding cut-off (0.10). In the structural model, agreeableness has a significantly positive effect on CQ in the German sample, whereas none of the path coefficients are significant in the Belgium sample.

To summarize, while the analysis of the German sample produces satisfactory levels of reliability, validity, and model fit, this is not the case with the Belgium sample. Specifically, both the measurement model results as well as model fit assessment would have clearly advised researchers against continuing the model interpretation with the Belgium sample. That is, researchers should drop such a sample (Belgian), report it transparently in the paper or the Appendix, and focus on the other country.²

Our analysis of the structural model results suggests that personality traits and international experience do not impact the CQ in the specific context of our study. With the exception of agreeableness, all path coefficient estimates are non-significant in the German sample. Although previous studies also found non-significant relationships between various personality traits and CQ, as well as between international experience and CQ, we recognize other potential reasons for these findings. Such non-significant relationships could be due to our use of student samples, to using an English questionnaire for non-native English speakers (e.g., Harzing, 2005; Schlägel & Sarstedt, 2016), and to our construct operationalizations. Specifically, using the Mini IPIP (Donnellan et al., 2006) and our composite measure of different biographical information related to international experience could have contributed to the model fit results. Short form measures of personality, such as the Mini IPIP, often focus on content validity while disregarding other psychometric properties such as reliability (e.g., Gosling et al., 2003). Future research should therefore take

² We nevertheless report the Belgium sample results in Table 9 for the sake of illustration.

this initial analysis as a starting point for further examining these relationships with larger non-student samples and different measures of personality traits and international experience.

5 Discussion

5.1 Findings, Recommendations and Conclusions

The empirical models used in IM/IB research to adequately represent its phenomena often involve both factors and components. The majority of SEM approaches are not able to handle factors and components in the same model; therefore, if used they result in biased parameter and structural estimates. In reviewing the use of different SEM approaches as reported in the six major journals we have shown that IM/IB researchers particularly use CSA and PLSPM without having adapted the two approaches that can handle factors and components – PLSc and IGSCA. IGSCA is particularly promising since it outperforms PLSc in terms of parameter recovery across a broad range of model specifications (Hwang et al., 2021).

Our simulation study shows that the GFI and SRMR indexes effectively discriminate between correct and misspecified models in IGSCA applications. Specifically, if the sample size increases, the GFI and SRMR consistently converge to their optimal values in the correct model, whereas they tend to differ from their optimal values in a misspecified model. Further, our results suggest that GFI's and SRMR's cut-off criteria are different to the GSCA's. This is not unexpected, as IGSCA extends to models which comprise both components and factors. As in GSCA (Cho et al., 2020), the selection of GFI's and SRMR's cut-off criteria for IGSCA depends on the sample size:

- If $N=50$, a GFI cut-off value of 0.84 and an SRMR cut-off value of 0.13 can be used to indicate an acceptable level of model fit. However, using the SRMR cut-off over the GFI cut-off is recommended because on average the former leads to much smaller Type 1 and II error rates than the latter.
- If $N=100$, a GFI cut-off value of 0.88 and an SRMR cut-off value of 0.10 can be chosen to indicate an acceptable level of model fit. If $N=200$, a GFI cut-off value of 0.91 and an SRMR cut-off value of 0.08 can be chosen. Although both indexes can be used to assess model fit in these sample sizes, using the suggested SRMR cut-off values might still be preferable to using the GFI cut-off values because the former generally results in smaller average Type I and Type II error rates. If the model's SRMR value is smaller than the suggested SRMR cut-off value, then a $GFI \geq 0.80$ might still indicate an acceptable fit for the sample sizes.
- If $N > 500$, a GFI cut-off value of 0.95 and an SRMR cut-off value of 0.06 can be chosen. In this case, one index is not preferred to the other, nor is a combination of the indexes preferred to using them separately. Each index's suggested cut-off value can be used independently to assess the model fit. That is, a $GFI \geq 0.95$ or an $SRMR \leq 0.06$ can indicate an acceptable fit.

Methodological overviews of samples size and statistical power (Ellis, 2010; Richter et al., 2016; Zhan, 2013) as well as recent meta-analyses in the IM/IB field (e.g., Christopoulou et al., 2021; Ding et al., 2021; Giachetti et al., 2019; Schlaegel et al., 2021) indicate that few studies use samples smaller than 100 and that IM/IB research is characterized by moderate sample sizes (N of about 250), with a significant number of studies using sample sizes of 100 to 150 observations. Our literature review of SEM articles in the main IM/IB journals across the last decade confirms this, showing that around a half of SEM studies use samples of less than 200 (see Table 2). For the few IM/IB studies that use samples sizes below 100, we recommend using the SRMR and a cut-off value 0.13. For the majority of IM/IB studies that use samples between 100 and 200, we also recommend the SRMR and a cut-off value of 0.10 in the assessment of IGSCA, which can be complemented by the GFI and a cut-off value of 0.88. The few IM/IB studies with rather large sample sizes (i.e., > 500) can use both the SRMR and the GFI with more conservative cut-off values (i.e., $GFI \geq 0.95$; $SRMR \leq 0.06$) and without a specific recommendation for one of the indices. In general, in evaluating fit indexes IM/IB researchers should use the cut-off values best suited to their respective sample sizes. The results of our simulation show that fit indices in IGSCA enable researchers to compare different models and to evaluate which model better approximates the reality and rule out alternative explanations.

Figure 6 presents a decision tree for applying IGSCA in IM/IB studies. First, researchers should carefully evaluate whether a model includes both components and common factors. If a model includes either factors *or* components, researchers should select one of the existing SEM approaches that suits the respective construct conceptualization (i.e., CSA, GSCA, or PLSPM). If the model includes both factors and components, then researchers should use IGSCA. Free software packages are available to analyze such models (e.g., <https://www.gscapro.com>). Based on the sample size and or specific respective recommendations for the use of SRMR and GFI as well as the significance and strength of relationships etc., researchers should carefully evaluate the model fit of IGSCA. If the model fit is not adequate, researchers should respecify the model based on theoretical considerations and justification as well as model testing results. If the model shows an adequate fit researcher should analyze and report the IGSCA results.

Our empirical application using a model that examines the impact of personality traits and international experience on CQ, a well-known concept from international management research, illustrates the efficacy of the indexes. Specifically, we find that the indexes indicate a low model fit in one of the two samples (Belgium), suggesting that the model isn't applicable to this specific international context. In fact, prior research already casts doubt on the CQ scale's universal applicability across different cultures (Schlägel & Sarstedt, 2016). Our results – while only used illustratively – offer further evidence for this skepticism.

5.2 Limitations and Future Research Directions

In this study, we reviewed the use of different SEM approaches in IM/IB research, highlighted the importance of factors and components in SEM and the usefulness

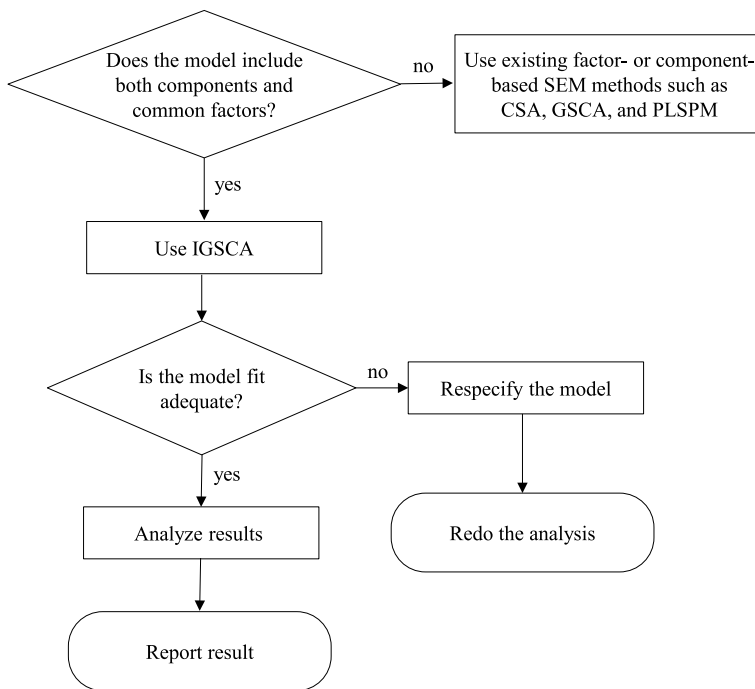


Fig. 6 Flow chart of decisions on the use of IGSCA

of IGSCA, systematically investigated how the GFI and SRMR indexes perform by analyzing simulated data under different experimental conditions, and provided an empirical application. This contributes to establishing an empirical basis for using IGSCA and these indices to evaluate models in IM/IB studies with both factors and components.

As with other simulation studies, our study has limitations. Although the population models used in our study are formulated based on prior studies that evaluated model fit indices in structural equation modeling (e.g., Cho et al., 2020; Hu & Bentler, 1999; Marsh et al., 2004; Sivo et al., 2006), they can be further expanded, for example, by accommodating a greater variety of both measurement and structural model specifications (e.g., Kingsley et al., 2017; Li et al., 2021; Lindner et al., 2020). Moreover, it is noteworthy that GFI and SRMR evaluate the discrepancy between sample and implied covariance matrices under the assumption that measurement errors are correlated within each block (Cho et al., 2021). Thus, it would be necessary to investigate whether our findings and recommendations remain valid when this assumption is violated. Accordingly, researchers should not rely solely on the suggested cut-off criteria of GFI and SRMR for model evaluation in IGSCA, but rather consider them reference points across model set-ups given the sample size (Niemand & Mai, 2018).

Also, a model's acceptable fit for a sample of observations does not necessarily guarantee that it also has sufficient predictive power for unseen observations, which

is highly recommended for ensuring the model's generalizability beyond the sample at hand (Cho et al., 2022; Sarstedt and Danks 2021; Liengaard et al., 2020; Shmueli, 2010). Further research should explore the combined use of model fit criteria and predictive power metrics (Cho et al., 2019) in developing guidelines for model assessment that combines these two evaluation perspectives.

Future research should also assess whether the GFI and SRMR cut-off values commonly proposed in a factor-based SEM context (e.g., Hu & Bentler, 1999; McDonald & Ho, 2002) can readily be transferred to $GSCA_M$. Since $GSCA_M$ reliably estimates models with factors only (Hwang et al., 2017, 2021), a $GFI \geq 0.90$ and an $SRMR \leq 0.08$ can be used to indicate an acceptable fit, but future research should substantiate these cut-off values for differently sized samples.

Finally, with reference to our empirical illustration, future research should assess the efficacy of different indicator weighting methods for the specification and estimation of higher-order component in an IGSCA context. While corresponding guidelines exist in a PLSPM context (Sarstedt et al. 2019; Wetzels et al. 2009), this is not the case with IGSCA.

Appendix

Appendix 1: Model Specification and Estimation in IGSCA

Here we provide a brief description of IGSCA's model specification and estimation based on Hwang et al. (2021). IGSCA involves specification of three sub-models: measurement, structural, and weighted relation models. The measurement model specifies the relationships between indicators and constructs that are represented by either factors or components, whereas the structural model expresses the relationships between constructs. The weighted relation model is used to specify a construct as a component with or without measurement errors explicitly incorporated. Let $\mathbf{z}_1$ and $\mathbf{z}_2$ denote vectors of J_1 composite indicators and J_2 effect indicators, respectively. Let $\boldsymbol{\gamma}_1$ and $\boldsymbol{\gamma}_2$ denote vectors of P_1 and P_2 constructs associated with $\mathbf{z}_1$ and $\mathbf{z}_2$, respectively. Assume that all indicators and constructs are normalized. Let $\mathbf{C}_1$ and $\mathbf{C}_2$ denote J_1 by P_1 and J_2 by P_2 matrices of loadings relating $\boldsymbol{\gamma}_1$ and $\boldsymbol{\gamma}_2$ to $\mathbf{z}_1$ and $\mathbf{z}_2$, respectively. Let $\mathbf{u}$ denote a vector of J_2 unique variables for $\mathbf{z}_2$. Let $\mathbf{D}$ denote a diagonal matrix of J_2 unique loadings for $\mathbf{z}_2$.

The measurement model for IGSCA is a combination of those used in GSCA and $GSCA_M$. The measurement model for GSCA is given as

$$\mathbf{z}_1 = \mathbf{C}_1 \boldsymbol{\gamma}_1 + \boldsymbol{\varepsilon}_1 \quad (\text{A-1})$$

where $\boldsymbol{\varepsilon}_1$ is the error term that represents the portion of $\mathbf{z}_1$ left unexplained by $\boldsymbol{\gamma}_1$ (Hwang & Takane, 2014, Chapter 2). The measurement model for $GSCA_M$ is given as

$$\mathbf{z}_2 = \mathbf{C}_2 \boldsymbol{\gamma}_2 + \mathbf{D} \mathbf{u} + \boldsymbol{\varepsilon}_2, \quad (\text{A-2})$$

where $\mathbf{C}_2\boldsymbol{\gamma}_2$ and $\mathbf{D}\mathbf{u}$ indicate common and unique parts of $\mathbf{z}_2$, respectively, and $\boldsymbol{\varepsilon}_2$ is the portion of $\mathbf{z}_2$ left unexplained by their common and unique parts (Hwang et al., 2017). We assume that $\boldsymbol{\gamma}_2$ is uncorrelated with $\mathbf{u}$ (i.e., $\boldsymbol{\gamma}_2\mathbf{u}' = \mathbf{u}\boldsymbol{\gamma}_2' = \mathbf{0}$) and $\mathbf{u}$ is column-wise orthonormalized (i.e., $\mathbf{u}\mathbf{u}' = \mathbf{I}_{J_2}$, where $\mathbf{I}_{J_2}$ is the identity matrix of order J_2).

The measurement model for IGSCA is then given as

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} \mathbf{C}_1; \mathbf{0} \\ \mathbf{0}; \mathbf{C}_2 \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{D}\mathbf{u} \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix} \quad (\text{A-3})$$

$$\mathbf{z} = \mathbf{C}\boldsymbol{\gamma} + \mathbf{s} + \boldsymbol{\varepsilon},$$

where $\mathbf{z} = [\mathbf{z}_1; \mathbf{z}_2]$, $\mathbf{C} = \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_2 \end{bmatrix}$, $\boldsymbol{\gamma} = [\boldsymbol{\gamma}_1; \boldsymbol{\gamma}_2]$, $\mathbf{s} = [\mathbf{0}; \mathbf{D}\mathbf{u}]$, and $\boldsymbol{\varepsilon} = [\boldsymbol{\varepsilon}_1; \boldsymbol{\varepsilon}_2]$.

The weighted relation model for IGSCA is also a combination of those for GSCA and GSCA_M. The weighted relation model for GSCA is given as

$$\boldsymbol{\gamma}_1 = \mathbf{W}_1\mathbf{z}_1, \quad (\text{A-4})$$

where $\mathbf{W}_1$ is a P_1 by J_1 matrix of weights assigned to $\mathbf{z}_1$ (Hwang & Takane, 2014, Chapter 2). The weighted relation model for GSCA_M is given as

$$\boldsymbol{\gamma}_2 = \mathbf{W}_2(\mathbf{z}_2 - \mathbf{D}\mathbf{u}), \quad (\text{A-5})$$

where $\mathbf{W}_2$ is a P_2 by J_2 matrix of weights assigned to $\mathbf{z}_2$, of which the unique parts are removed (Hwang et al., 2017). Then, the weighted relation model for IGSCA is given as

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} = \begin{bmatrix} \mathbf{W}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_2 \end{bmatrix} \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 - \mathbf{D}\mathbf{u} \end{bmatrix} \quad (\text{A-6})$$

$$\boldsymbol{\gamma} = \mathbf{W}(\mathbf{z} - \mathbf{s}),$$

where $\mathbf{W} = \begin{bmatrix} \mathbf{W}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_2 \end{bmatrix}$. This sub-model shows that both $\boldsymbol{\gamma}_1$ and $\boldsymbol{\gamma}_2$ are defined as components rather than factors. Nonetheless, as (A-5) shows, $\boldsymbol{\gamma}_2$ represents components of effect indicators, of which the unique parts are removed. In this way, measurement errors in the indicators are taken into account, meaning the parameters involving $\boldsymbol{\gamma}_2$ are similar to those in factor-based SEM (Hwang et al., 2017).

Let $\mathbf{B}$ denote a P by P matrix of path coefficients relating $\boldsymbol{\gamma}$ among themselves, where $P = P_1 + P_2$. The structural model for IGSCA is expressed as

$$\boldsymbol{\gamma} = \mathbf{B}\boldsymbol{\gamma} + \boldsymbol{\zeta}, \quad (\text{A-7})$$

where $\boldsymbol{\zeta}$ is the error term for $\boldsymbol{\gamma}$. The $\mathbf{B}$ matrix contains path coefficients relating components to factors, as well as those among either factors or components only.

Our approach integrates the sub-models into a unified formulation, referred to as the IGSCA model, as follows:

$$\begin{bmatrix} \mathbf{z} \\ \boldsymbol{\gamma} \end{bmatrix} = \begin{bmatrix} \mathbf{C} \\ \mathbf{B} \end{bmatrix} \boldsymbol{\gamma} + \begin{bmatrix} \mathbf{s} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\varepsilon} \\ \boldsymbol{\zeta} \end{bmatrix} \quad (\text{A-8})$$

$$\boldsymbol{\Psi} = \mathbf{A}\boldsymbol{\gamma} + \mathbf{v} + \mathbf{e},$$

where $\boldsymbol{\Psi} = [\mathbf{z}; \boldsymbol{\gamma}]$, $\mathbf{A} = [\mathbf{C}; \mathbf{B}]$, $\mathbf{v} = [\mathbf{s}; \mathbf{0}]$, and $\mathbf{e} = [\boldsymbol{\varepsilon}; \boldsymbol{\zeta}]$. As in GSCA, IGSCA does not make a distributional assumption of $\mathbf{e}$, i.e., multivariate normality.

Let $\mathbf{e}_i$ denote the error term in (A-8) for a single observation of a sample of N observations ($i = 1, \dots, N$). To estimate all parameters in $\mathbf{W}$, $\mathbf{A}$ and $\mathbf{v}$, IGSCA aims to minimize the following objective function

$$\varphi = \sum_{i=1}^N \mathbf{e}_i' \mathbf{e}_i \quad (\text{A-9})$$

subject to $\text{diag}(\boldsymbol{\gamma}\boldsymbol{\gamma}') = \mathbf{I}_p$, $\boldsymbol{\gamma}_2\mathbf{u}' = \mathbf{u}\boldsymbol{\gamma}_2' = \mathbf{0}$, and $\mathbf{u}\mathbf{u}' = \mathbf{I}_{J_2}$, where $\text{diag}()$ denotes a diagonal matrix. An iterative algorithm is used to minimize the objective function by a simple adoption of the existing algorithms for GSCA and GSCA_M (Hwang et al., 2021). This algorithm repeats several steps, each of which updates one set of parameters while the other sets are fixed, until no substantial differences in parameter estimates occur between iterations. Specifically, in step 1, the unique variables $\mathbf{u}$ and unique loadings $\mathbf{D}$ are updated. In step 2, the component weights $\mathbf{W}$ are updated using the GSCA algorithm for $\mathbf{W}_1$ and the GSCA_M algorithm for $\mathbf{W}_2$. In the next steps, the loadings $\mathbf{C}$ and path coefficients $\mathbf{B}$ are estimated.

Appendix 2: Model Fit Indexes

IGSCA offers two overall fit indexes, GFI and SRMR, which summarize the size of the differences between the sample and model-implied covariances. The implied covariances obtained from the parameter estimates on convergence are also called reproduced covariances. Although in principle, IGSCA does not assume indicators have a specific covariance structure to derive the reproduced covariances, we assume that composite indicators per component are correlated to each other, leading the covariance matrix of $\boldsymbol{\varepsilon}_1$ to be block-diagonal, as is typically assumed in component-based SEM (e.g., Bollen & Bauldry, 2011; Cho & Choi, 2020; Dijkstra, 2017; Grace & Bollen, 2008). Also, we assume that the common factor analytic model holds for a sample, meaning $\boldsymbol{\varepsilon}_2$ will be zero (Velicer & Jackson, 1990). Further, we assume that $\text{cov}(\mathbf{v}, \mathbf{e}) = \mathbf{0}$ and $\text{cov}(\boldsymbol{\varepsilon}, \boldsymbol{\zeta}) = \mathbf{0}$. Then, we can re-express (A-8) as

$$\begin{bmatrix} \mathbf{I}_J \\ \mathbf{W} \end{bmatrix} (\mathbf{z} - \mathbf{s}) = \begin{bmatrix} \mathbf{0} & \mathbf{C} \\ \mathbf{0} & \mathbf{B} \end{bmatrix} \begin{bmatrix} \mathbf{I}_J \\ \mathbf{W} \end{bmatrix} (\mathbf{z} - \mathbf{s}) + \begin{bmatrix} \boldsymbol{\varepsilon} \\ \boldsymbol{\zeta} \end{bmatrix} \quad (\text{A-10})$$

$$\mathbf{V}(\mathbf{z} - \mathbf{s}) = \mathbf{T}\mathbf{V} \begin{bmatrix} \mathbf{I}_J \\ \mathbf{W} \end{bmatrix} (\mathbf{z} - \mathbf{s}) + \mathbf{e}$$

$$\mathbf{V}(\mathbf{z} - \mathbf{s}) = (\mathbf{I}_K - \mathbf{T})^{-1} \mathbf{e},$$

where $J = J_1 + J_2$, $K = J + P$, and $\mathbf{T} = \begin{bmatrix} \mathbf{0} & \mathbf{C} \\ \mathbf{0} & \mathbf{B} \end{bmatrix}$. The implied covariance matrix of $\mathbf{z}$, denoted by $\boldsymbol{\Sigma}$, is then given as

$$\Sigma = \mathbf{G}(\mathbf{I}_K - \mathbf{T})^{-1} \mathbf{E}(\mathbf{e}\mathbf{e}')((\mathbf{I}_K - \mathbf{T})^{-1})' \mathbf{G}' + \mathbf{E}(\mathbf{s}\mathbf{s}'), \quad (\text{A-11})$$

where $\mathbf{G} = [\mathbf{I}_J, \mathbf{0}]$, $\mathbf{E}(\mathbf{e}\mathbf{e}')$ is a block-diagonal covariance matrix of $\mathbf{e}$, and $\mathbf{E}(\mathbf{s}\mathbf{s}')$ is a diagonal covariance matrix of $\mathbf{s}$, of which the non-zero entries are equal to $\mathbf{D}^2$.

Let $\mathbf{S}$ and $\hat{\Sigma}$ denote the sample and the reproduced covariance matrices. Let s_{ij} and $\hat{\sigma}_{ij}$ respectively denote an observed covariance in $\mathbf{S}$ and the corresponding reproduced covariance in $\hat{\Sigma}$. Then, the GFI and SRMR are calculated as

$$\text{GFI} = 1 - \frac{\text{trace}([\mathbf{S} - \hat{\Sigma}]^2)}{\text{trace}(\mathbf{S}^2)}, \text{ and} \quad (\text{A-12})$$

$$\text{SRMR} = \sqrt{\left\{ 2 \sum_{j=1}^J \sum_{l=1}^j [(s_{jl} - \hat{\sigma}_{jl}) / (s_{jj}s_{ll})]^2 \right\} / J(J+1)} \quad (\text{A-13})$$

As the above formulas show, GFI values close to 1 and SRMR values close to 0 indicate a small degree of the covariance discrepancies.

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Data availability In the case of article acceptance in an open-source repository.

Code availability In the case of article acceptance in an open-source repository.

Declarations

Conflicts of interest Not applicable.

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